the calculation in motion. In a couple of seconds the computer returned its answer: 8.999999. My heart sank. Could it be that space-tearing flop transitions shatter the mirror relation, likely indicating that they cannot actually occur? Almost immediately, though, we all realized that something funny must be going on. If there was a real mismatch in the physics following from the two shapes, it was extremely unlikely that the computer calculation should yield an answer so close to a whole number. If our ideas were wrong, there was no reason in the world to expect anything but a random collection of digits. We had gotten a wrong answer, but one that suggested, perhaps, that we had just made some simple arithmetic error. Aspinwall and I went to the blackboard, and in a moment we found our mistake: we had dropped a factor of 3 in the "simpler" calculation we had done weeks before; the true result was 9. The computer answer was therefore just what we wanted.

Of course, the after-the-fact agreement was only marginally convincing. When you know the answer you want, it is often all too easy to figure out a way of getting it. We needed to do another example. Having already written all of the necessary computer code, this was not hard to do. We calculated another particle mass on the upper Calabi-Yau shape, being careful this time to make no errors. We found the answer: 12. Once again, we huddled around the computer and set it on its way. Seconds later it returned 11.999999. Agreement. We had shown that the supposed mirror is the mirror, and hence space-tearing flop transitions are part of the physics of string theory.

At this I jumped out of my chair and ran an unrestrained victory lap around the office. Morrison beamed from behind the computer. Aspinwall's reaction, though, was rather different. "That's great, but I knew it would work," he calmly said. "And where's my beer?"

Witten's Approach

That Monday, we triumphantly went to Witten and told him of our success. He was very pleased with our result. And, as it turned out, he too had just found a way of establishing that flop transitions occur in string theory. His argument was quite different from ours, and it significantly illuminates the microscopic understanding of why the spatial tears do not have any catastrophic consequences.